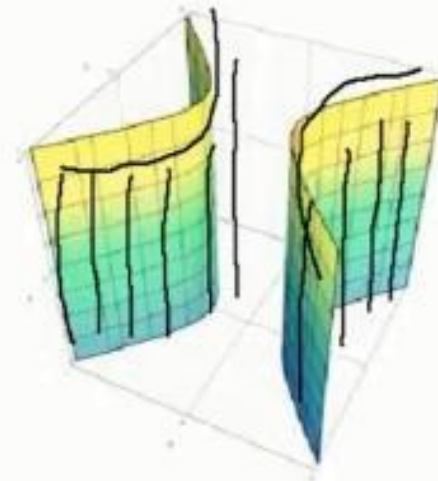


Hyperbolic Cylinder

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{a^2} - \frac{z^2}{b^2} = 1$$

$$\frac{y^2}{a^2} - \frac{z^2}{b^2} = 1$$



Example. Discuss the surface $5x^2 - 12y^2 - 7 = 0$

Sol.

$$5x^2 - 12y^2 - 7 = 0$$

$$5x^2 - 12y^2 = 7$$

$$\frac{5x^2}{7} - \frac{12y^2}{7} = 1$$

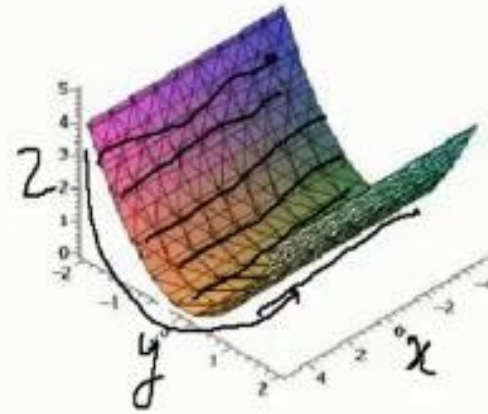
$$\frac{x^2}{7/5} - \frac{y^2}{7/12} = 1 \Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Parabolic Cylinder

$$y = cz^2$$

$$y = -5x^2$$

$$x = cz^2$$

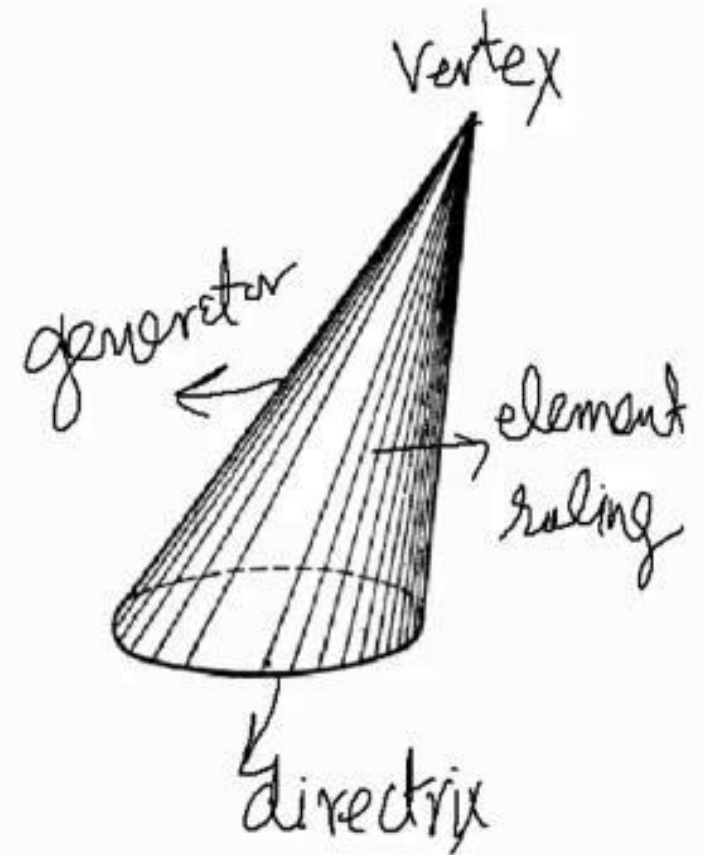
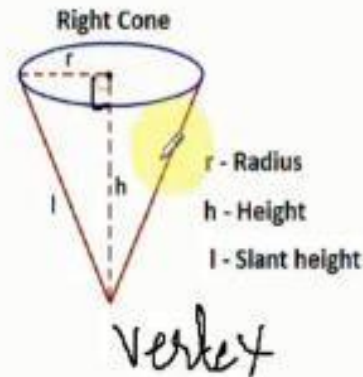


Cone

A surface generated by a straight line passing through a fixed point and intersecting a fixed plane curve not containing the fixed point is called a cone. The given curve is called the directrix and the fixed point is called the vertex of the cone. The straight line is called the generator of and any particular position of the generator is called an element or ruling of the cone.

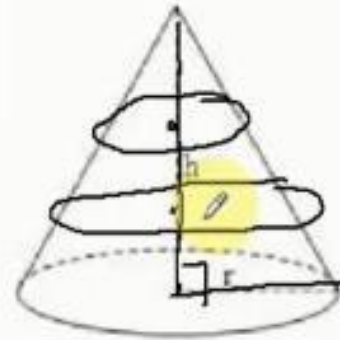
Right cone

A right cone is a cone with its vertex above the centre of its base.



Right circular cone

A circular cone is right circular if the centres of its sections form a line perpendicular to its base.



Example Find an eq. of cone whose directrix is

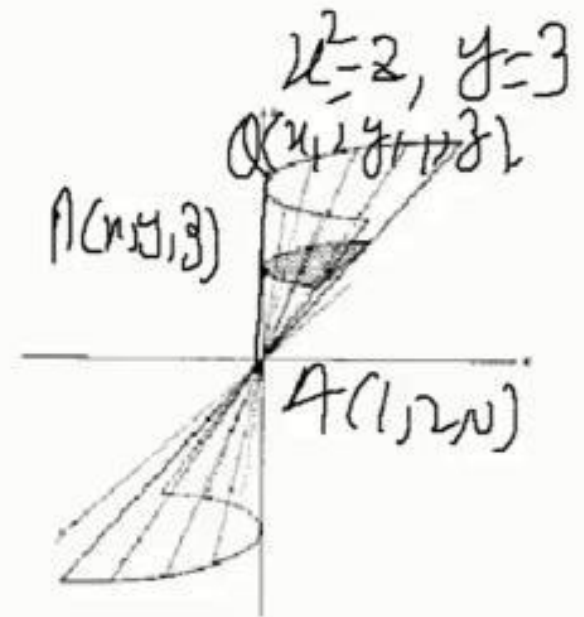
$$x^2 = z, \quad y = 3 \quad \text{--- (1)}$$

and whose vertex is $A(1, 2, 0)$.

Sol $x_1^2 = z_1, \quad y_1 = 3 \quad \text{--- (2)}$

$$\frac{x-1}{x_1-1} = \frac{y-2}{y_1-2} = \frac{z-0}{z_1-0} \quad \text{--- (3)}$$

$$\frac{x-1}{x_1-1} = \frac{y-2}{y_1-2} \Rightarrow \frac{x-1}{x_1-1} = \frac{y-2}{3-2} = y-2$$



$$\frac{x-1}{x_1-1} = \frac{y-2}{y_1-2} \Rightarrow \frac{x-1}{x_1-1} = \frac{y-2}{3-2} = y-2$$

$$x-1 = (x_1-1)(y-2)$$

$$\Rightarrow x_1 - 1 = \frac{x-1}{y-2} \Rightarrow x_1 = \frac{x-1}{y-2} + 1 = \frac{x-1+y-2}{y-2}$$

$$= \frac{x+y-3}{y-2}$$

$$\Rightarrow x_1 = \frac{x+y-3}{y-2}$$

$$\frac{z}{z_1} = \frac{y-2}{y_1-2} \Rightarrow \frac{z}{z_1} = y-2,$$

$$\Rightarrow z_1 = \frac{z}{y-2}$$

$$\frac{z}{z_1} = \frac{y-2}{y_1-2} \Rightarrow \frac{z}{z_1} = y-2$$

$$\Rightarrow z_1 = \frac{z}{y-2}$$

$$\left(\frac{x+y-3}{y-2} \right)^2 = \frac{z}{y-2}$$

$$(x+y-3)^2 = \left(\frac{z}{y-2} \right) (y-2)^2$$

$$(x+y-3) = z(y-2)$$

$$x^2 + y^2 + 2xy + 9 - 6x - 6y = zy - 2z$$

$$x^2 + y^2 + 2xy - 6x - 6y + 2z - yz + 9 = 0$$

which is eq. of Cone

x-intercept $y=0, z=0$

$$x^2 - 6x + 9 = 0 \Rightarrow (x-3)^2 = 0 \Rightarrow x=3$$

y-intercept $x=0, z=0$

$$y^2 - 6y + 9 = 0 \Rightarrow (y-3)^2 = 0 \Rightarrow y=3$$

z-intercept $x=0, y=0$

$$2z + 9 = 0 \Rightarrow z = -9/2$$

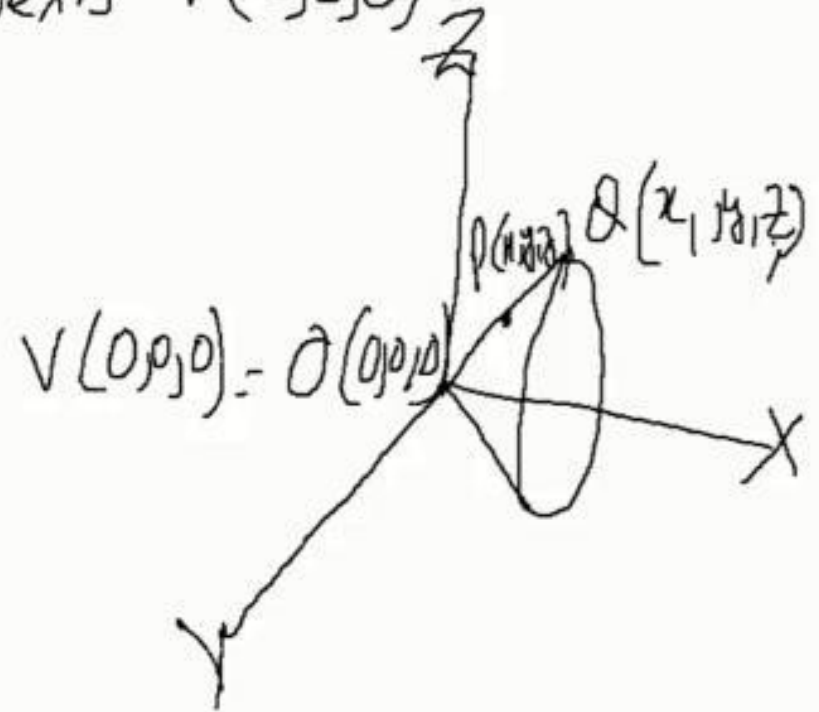
Example. Find an eq. of the cone whose directrix is
 $y^2 + z^2 = 1$; $x = 2$ and vertex is $V(0,0,0)$

Sol.

$$y_1^2 + z_1^2 = 1 ; x_1 = 2 \quad \text{--- (1)}$$

$$\frac{x-0}{x_1-0} = \frac{y-0}{y_1-0} = \frac{z-0}{z_1-0}$$

$$\Rightarrow \frac{x}{x_1} = \frac{y}{y_1} = \frac{z}{z_1} \quad \text{and} \quad \frac{y}{y_1} = \frac{z}{z_1}$$



$$\therefore x_1 = 2$$

$$\frac{x}{2} = \frac{y}{y_1} \quad \text{and}$$

$$\frac{x}{2} = \frac{z}{z_1}$$

$$\Rightarrow y_1 = \frac{2y}{x}$$

$$z_1 = \frac{2z}{x}$$

$$\left(\frac{2y}{x}\right)^2 + \left(\frac{2z}{x}\right)^2 = 1$$

$$\frac{4y^2}{x^2} + \frac{4z^2}{x^2} = 1$$

$$4y^2 + 4z^2 = x^2$$

which is eq. of cone

Elliptic Cone

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = c z^2 \quad c > 0$$

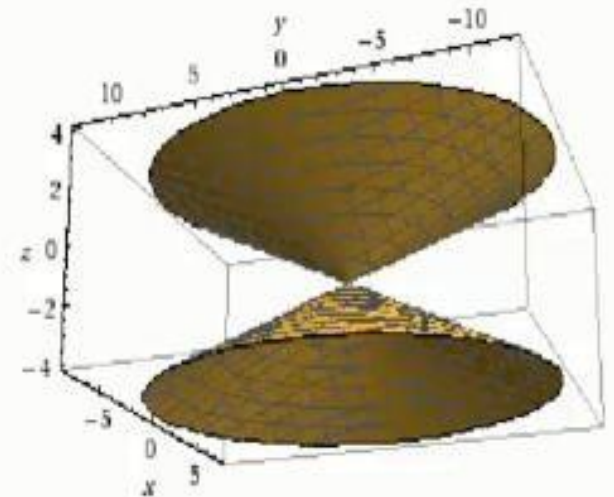
→ along z-axis

if $c=1$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z^2$$

$$\frac{x^2}{a^2} + \frac{z^2}{b^2} = y^2 \Rightarrow \text{along y-axis}$$

$$\frac{y^2}{a^2} + \frac{z^2}{b^2} = x^2 \Rightarrow \text{along x-axis}$$



Properties

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z^2$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - z^2 = 0 \Rightarrow f(x, y, z) = 0$$

Symmetry

(i) xy-plane $f(x, y, z) = 0 \Rightarrow f(x, y, -z) = 0$

$$f(x, y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - (-z)^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2} - z^2 = 0 \Rightarrow f(x, y, z)$$

$\Rightarrow$ Symmetrical about xy-plane

Act
Go t

Intercepts $f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - z^2 = 0$

x-intercept $y=0, z=0$

$$\frac{x^2}{a^2} = 0 \Rightarrow x=0$$

y-intercept $x=0, z=0 \Rightarrow y=0$

z-intercept $x=0, y=0 \Rightarrow z=0$
• $O(0, 0, 0)$

Traces $f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - z^2 = 0$

(i) xy-plane $z=0 \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 0 \Rightarrow$ represents a pt. $(0, 0, 0)$

(ii) xz-plane $y=0$
 $\frac{x^2}{a^2} - z^2 = 0 \Rightarrow \frac{x^2}{a^2} = z^2 \Rightarrow x^2 = a^2 z^2$

$\Rightarrow x = \pm az$

(iii) yz-plane $x=0$
 $\frac{y^2}{b^2} - z^2 = 0 \Rightarrow y^2 = b^2 z^2$
 $\Rightarrow y = \pm bz$

Ellipsoid

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} + \frac{(z-l)^2}{c^2} = 1 \quad (1)$$

Centre (h, k, l)

Centre $(0, 0, 0)$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

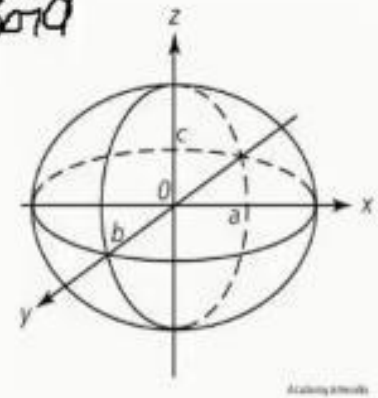
If $a = b = c$

$$\frac{x^2}{a^2} + \frac{y^2}{a^2} + \frac{z^2}{a^2} = 1$$

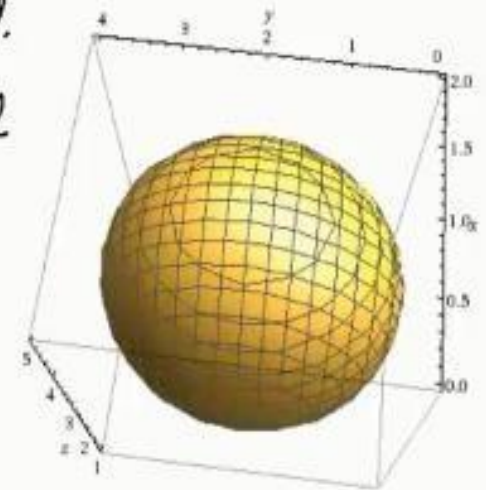
$$x^2 + y^2 + z^2 = a^2$$

$C(0,0,0) \quad r=a$

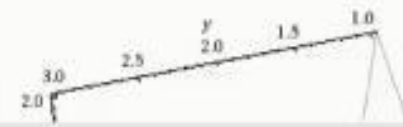
Ellipsoid



Spheroid.
oblate



Spheroid

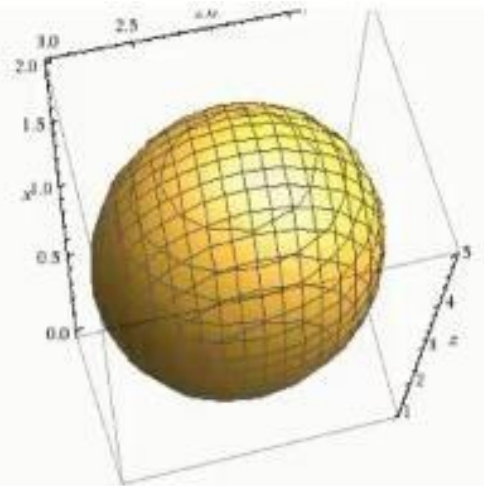


$$C(0,0,0) \quad r=a$$

Spheroid

$$a=b, \quad c \neq a, \quad c \neq b$$

Spheroid
prolate



if $a=b$ $c < a, c < b \rightarrow$ oblate

if $a=b$ $c > a, c > b \rightarrow$ Prolate

Act

Properties

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Symmetry

(i) xy-plane $f(x, y, z) = f(x, y, -z) = 0$

$$f(x, y, -z) = 0 \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{(-z)^2}{c^2} = 1$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \Rightarrow f(x, y, z) = 0$$

$\Rightarrow$ Symmetrical about xy-plane

(ii) yz-plane

$$f(x, -y, z) = 0 \Rightarrow f(x, y, z) = 0$$

(iii) xz-plane $f(x, y, z) = 0 \Rightarrow f(x, y, z) = 0$

$$f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$\Rightarrow f(x, y, z) = 0$$

$\Rightarrow$ Symmetrical about xz-plane

(iv) Similarly

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$\Rightarrow f(x, y, z) = 0$$

$\Rightarrow$ Symmetrical about xz -plane

(ii) Similarly symmetrical about yz -plane

(iv) Symmetrical about all coordinate axes
as well as about origin

Traces $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

(i) xy-plane

$$z=0$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$\rightarrow$ ellipse

(ii) xz-plane

$$y=0$$

$$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1$$

$\Rightarrow$ ellipse

(iii) yz-plane

$$x=0$$

$$\Rightarrow \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Intercepts

(i) x-intercept Put $y=0, z=0$

$$\frac{x^2}{a^2} = 1 \Rightarrow x^2 = a^2 \Rightarrow x = \pm a$$

$(a, 0, 0)$ and $(-a, 0, 0)$

(i') y-intercept Put $x=0, z=0$

$$\frac{y^2}{b^2} = 1 \Rightarrow y^2 = b^2 \Rightarrow y = \pm b$$

$(0, b, 0)$ and $(0, -b, 0)$

Act

(ii) y-intercept Put $x=0, z=0$

$$\frac{y^2}{b^2} = 1 \Rightarrow y^2 = b^2 \Rightarrow y = \pm b$$

$(0, b, 0)$ and $(0, -b, 0)$

(iii) z-intercept $x=0, y=0$

$$\frac{z^2}{c^2} = 1 \Rightarrow z^2 = c^2 \Rightarrow z = \pm c$$

$(0, 0, c)$ and $(0, 0, -c)$

Act

Example Identify the surface

$$100x^2 + 25y^2 + 4z^2 - 400x - 150y - 8z + 529 = 0$$

and find its traces and intercepts

Solution,

$$100x^2 + 25y^2 + 4z^2 - 400x - 150y - 8z + 529 = 0$$

$$100x^2 - 400x + 25y^2 - 150y + 4z^2 - 8z = -529$$

$$100(x^2 - 4x) + 25(y^2 - 6y) + 4(z^2 - 2z) = -529$$

$$100(x^2 - 4x + 4) + 25(y^2 - 6y + 9) + 4(z^2 - 2z + 1) = -529 + 400 + 225 + 4$$

$$100(x^2 - 4x + 4) + 25(y^2 - 6y + 9) + 4(z^2 - 2z + 1) = -529$$

$$100(x-2)^2 + 25(y-3)^2 + 4(z-1)^2 = -529 + 400 + 225 + 4$$

$$100(x-2)^2 + 25(y-3)^2 + 4(z-1)^2 = 100$$

$$\frac{(x-2)^2}{1} + \frac{(y-3)^2}{4} + \frac{(z-1)^2}{25} = 1$$

$$\frac{(x-2)^2}{1^2} + \frac{(y-3)^2}{2^2} + \frac{(z-1)^2}{5^2} = 1$$

$$\frac{(x-2)^2}{1^2} + \frac{(y-3)^2}{2^2} + \frac{(z-1)^2}{5^2} = 1$$

which is of the form

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} + \frac{(z-l)^2}{c^2} = 1$$

∴ The given eq. is the eq. of ellipsoid.

C(2, 3, 1)

Act
Go t

Traces

(1) xy-plane $z=0$

$$(x-2)^2 + \frac{(y-3)^2}{4} + \frac{(0-1)^2}{25} = 1$$

$$(x-2)^2 + \frac{(y-3)^2}{4} + \frac{1}{25} = 1$$

$$(x-2)^2 + \frac{(y-3)^2}{4} = 1 - \frac{1}{25} = \frac{24}{25}$$

$$\frac{25(x-2)^2}{24} + \frac{25(y-3)^2}{4} = 1 \Rightarrow \text{ellipse}$$

(ii) yz-plane

$$x=0$$

$$(0-2)^2 + \frac{(y-3)^2}{4} + \frac{(z-1)^2}{25} = 1$$

$$4 + \frac{(y-3)^2}{4} + \frac{(z-1)^2}{25} = 1$$

$$\frac{(y-3)^2}{4} + \frac{(z-1)^2}{25} = -3 \Rightarrow \text{No trace}$$

(iii) xz-plane

$$y=0$$

$$(x-2)^2 + \frac{(0-3)^2}{4} + \frac{(z-1)^2}{25} = 1$$

(iii) xz-plane $y=0$ $\Rightarrow$ No trace

$$(x-2)^2 + \frac{(0-3)^2}{4} + \frac{(z-1)^2}{25} = 1$$

$$(x-2)^2 + \frac{9}{4} + \frac{(z-1)^2}{25} = 1$$

$$(x-2)^2 + \frac{(z-1)^2}{25} = 1 - \frac{9}{4} = -\frac{5}{4}$$

$\Rightarrow$ No trace in xz-plane

Intercepts

(i) x-intercept put $y=0, z=0$

$$(x-2)^2 + \frac{(0-3)^2}{4} + \frac{(0-1)^2}{25} = 1$$

$$(x-2)^2 + \frac{9}{4} + \frac{1}{25} = 1 \Rightarrow (x-2)^2 = 1 - \frac{9}{4} - \frac{1}{25} < 0$$

$\Rightarrow$ No x-Intercept

y-intercept

$$x=0, z=0$$

$$\frac{(0-2)^2}{1} + \frac{(y-3)^2}{4} + \frac{(0-1)^2}{25} = 1$$

$$\frac{(y-3)^2}{4} + 4 + \frac{1}{25} = 1$$

$$\frac{(y-3)^2}{4} = 1 - 4 - \frac{1}{25} < 0 \Rightarrow$$

No y-intercept

z-intercept

$$x=0, y=0$$

$$\frac{(0-2)^2}{1} + \frac{(0-3)^2}{4} + \frac{(z-1)^2}{25} = 1$$

$$4 + \frac{9}{4} + \frac{(z-1)^2}{25} = 1$$

$$\frac{(z-1)^2}{25} = 1 - 4 - \frac{9}{4} < 0$$

$\Rightarrow$ No z-intercept.

Example Discuss the Symmetry of the ellipsoid.

$$x^2 + \frac{y^2}{9} + \frac{(z-1)^2}{4} = 1$$

Solution (i) xy-plane

$$f(x, y, z) = f(x, y, z) \Rightarrow$$

$$f(x, y, -z) = x^2 + \frac{y^2}{9} + \frac{(-z-1)^2}{4} = 1$$

$$= x^2 + \frac{y^2}{9} + \frac{(z+1)^2}{4} = 1 \neq f(x, y, -z)$$

$\Rightarrow$ not symmetrical about xy-plane

(ii) xz-plane

$$\begin{aligned} f(x, -y, z) &= x^2 + \frac{(-y)^2}{9} + \frac{(z-1)^2}{4} \\ &= x^2 + \frac{y^2}{9} + \frac{(z-1)^2}{4} \\ &= f(x, y, z) \end{aligned}$$

$\Rightarrow$ Symmetrical about xz-plane

iii) yz plane $f(-x, y, z) = (-x)^2 + \frac{y^2}{9} + \frac{(z-1)^2}{4}$
 $= x^2 + \frac{y^2}{9} + \frac{(z-1)^2}{4}$
 $= f(x, y, z)$

$\Rightarrow$ Symmetrical about yz-plane.

Elliptic paraboloid

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = c(z-l)$$

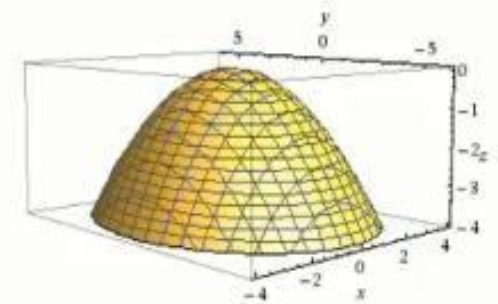
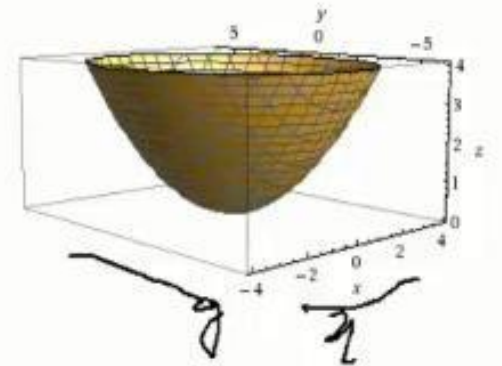
$V(h, k, l)$

$V(0, 0, 0)$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = cz$$

if $c > 0 \Rightarrow$ lies above xy -plane

if $c < 0 \Rightarrow$ lies below xy -plane



Activate
Go to Sett

e.g. $\frac{x^2}{4} + \frac{y^2}{9} = z, \quad c=1$

Open along z-axis above xy-plane

or $\frac{x^2}{4} + \frac{y^2}{9} = -z, \quad c=-1$

open along z-axis, lies below xy-plane

$\frac{x^2}{a^2} + \frac{z^2}{b^2} = y \Rightarrow$ open along y-axis

$\frac{y^2}{a^2} + \frac{z^2}{b^2} = x \Rightarrow$ open along x-axis

Activate V
Go to Setting

Symmetry

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z$$

$$f(x, y, z) = 0$$

$$f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0$$

(i) about xy-plane

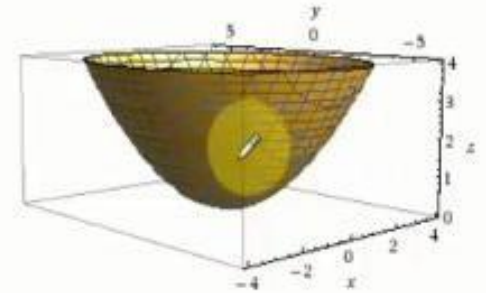
$$f(x, y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + z \neq 0$$

$\Rightarrow$ Not Symmetrical

(ii) xz-plane

$$f(x, -y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0$$

$\Rightarrow$ Symmetrical about xz-plane



Activate V
Go to Setting

(ii) about yz-plane

$$f(-x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0$$

$\Rightarrow$ Symmetrical about yz-plane

(iv) about x-axis

$$f(x, -y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + z = 0$$

$\Rightarrow$ not Symmetrical about x-axis

(v) about y-axis

$$f(-x, y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + z \neq 0$$

$\Rightarrow$ Not Symmetrical

(vi) about z-axis

$$f(-x, -y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - z = 0$$

$\Rightarrow$ Symmetrical

(vii) about origin

$$f(-x, -y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + z \neq 0$$

$\Rightarrow$ Not Symmetrical

Traces

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z$$

(i) xy-plane

$$z=0$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 0$$

$\Rightarrow$ which is possible only $x=0, y=0$

i.e. Trace is origin in xy-plane.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad z=1 \Rightarrow \text{ellipse}$$

Activate
Go to Setting

(i) yz-plane $x=0$

$$\frac{y^2}{b^2} = z \Rightarrow y^2 = b^2 z \Rightarrow \text{parabola}$$

(ii) xz-plane $y=0$

$$\frac{x^2}{a^2} = z \Rightarrow x^2 = a^2 z \Rightarrow \text{parabola}$$

Intercepts

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = z$$

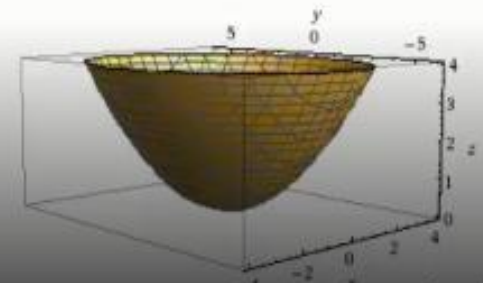
z-intercept

$$x=0, y=0, \Rightarrow z=0$$

i.e. only intercept is origin.

Boundedness

Not bounded



▶ 7:37 / 7:37



Activate V
Go to Setting

Example Elliptic paraboloid. Identify the surface

$25x^2 + y^2 - 2y - 25z + 1 = 0$ and find its traces in the coordinate planes and all intercepts.

Sol

$$25x^2 + y^2 - 2y - 25z + 1 = 0$$

$$25x^2 + y^2 - 2y + 1 = 25z$$

$$25x^2 + (y-1)^2 = 25z$$

$$x^2 + \frac{(y-1)^2}{25} = z$$

Comparing with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = z$

$\Rightarrow$ elliptic paraboloid

Activate V
Go to Setting

Traces

(i) xy-plane $z=0$

$$x^2 + \frac{(y-1)^2}{25} = 0$$

$$\Rightarrow x=0, \quad y-1=0 \text{ i.e. } y=1$$

$$\text{i.e. } (0, 1, 0)$$

(ii) yz-plane $x=0$

$$0^2 + \frac{(y-1)^2}{25} = z \Rightarrow (y-1)^2 = 25z \Rightarrow \text{parabola}$$

Intercepts

(i) x-intercept $y=0, z=0$

$$x^2 + \frac{(0-1)^2}{25} = 0 \Rightarrow x^2 + \frac{1}{25} = 0$$

$$\Rightarrow x^2 = -\frac{1}{25} \Rightarrow \text{impossible}$$

$\Rightarrow$ No-intercept with x-axis

(ii) y-intercept $x=0, z=0$

$$0 + \frac{(y-1)^2}{25} = 0 \Rightarrow (y-1)^2 = 0 \Rightarrow y-1=0 \Rightarrow y=1$$

$(0, 1, 0)$

(iii) z-intercept $x=0, y=0$

$$0 + \frac{1}{25} = z \Rightarrow z = \frac{1}{25}$$

$(0, 0, \frac{1}{25})$

Traces xy-plane put $z=0$

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 0 \Rightarrow \frac{y^2}{b^2} = \frac{x^2}{a^2} \Rightarrow y = \pm \frac{b}{a}x \Rightarrow \text{Pair of intersecting lines}$$

xz-plane put $y=0$

$$-\frac{x^2}{a^2} = z \Rightarrow x^2 = -a^2 z \Rightarrow \text{Parabola.}$$

yz-plane put $x=0$

$$\frac{y^2}{b^2} = z \Rightarrow y^2 = b^2 z \Rightarrow \text{Parabola}$$

put $z=1$

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1 \Rightarrow \text{Hyperbola}$$



Symmetry

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} - z = 0$$

$$f(x, y, z) = \frac{y^2}{b^2} - \frac{x^2}{a^2} - z = 0 \Rightarrow (2)$$

(i) xy-plane $f(x, y, z) = \frac{y^2}{b^2} - \frac{x^2}{a^2} + z \neq 0$ by (2).
 $\Rightarrow$ Not Symmetrical

(ii) yz-plane

$$f(-x, y, z) = \frac{y^2}{b^2} - \frac{(-x)^2}{a^2} - z = \frac{y^2}{b^2} - \frac{x^2}{a^2} - z = 0 \text{ by (2)} \\ \Rightarrow \text{Symmetrical}$$

Activate V
Go to Setting

$\Rightarrow$ symmetrical

(iii) xz-plane $f(x, -y, z) = \frac{y^2}{b^2} - \frac{x^2}{a^2} - z = 0$ by (i)

$\Rightarrow$ Symmetrical about xz-plane

(iv) x-axis $f(x, -y, -z) = \frac{y^2}{b^2} - \frac{x^2}{a^2} + z \neq 0 \Rightarrow$ Not Symmetrical

(v) y-axis $f(-x, y, -z) = \frac{y^2}{b^2} - \frac{x^2}{a^2} + z \neq 0 \Rightarrow$ Not Symmetrical

(vi) z-axis $f(-x, -y, z) = \frac{y^2}{b^2} - \frac{x^2}{a^2} - z = 0 \Rightarrow$ Symmetrical

(vii) origin $f(-x, -y, -z) = \frac{y^2}{b^2} - \frac{x^2}{a^2} + z \neq 0$ Not Symmetrical

Intercepts

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = z.$$

$$y=0, z=0$$

$$\frac{0^2}{b^2} - \frac{x^2}{a^2} = 0 \Rightarrow x^2 = 0 \Rightarrow x=0$$

$$x=0, y=0, z=0$$

$$O(0,0,0)$$

Example Identify the surface

$$-x^2 + 4y^2 - 4x - 16y - 4z + 12 = 0$$

and find its traces in the coordinate planes and intercepts.

Sol

$$-x^2 + 4y^2 - 4x - 16y - 4z + 12 = 0$$

$$4y^2 - 16y - x^2 - 4x + 12 = 4z$$

$$4(y^2 - 4y + 4) - (x^2 + 4x + 4) + 12 = 4z + 16 - 4$$

$$4(y - 2)^2 - (x + 2)^2 + 12 = 4z + 12$$

$$(y - 2)^2 - \frac{(x + 2)^2}{4} = z$$

Hyperbolic paraboloid

Activate V
Go to Setting

Traces $(y-2)^2 - \frac{(x+2)^2}{4} = z$

xy-plane put $z=0 \Rightarrow (y-2)^2 = \frac{(x+2)^2}{4}$

$$\Rightarrow y-2 = \pm \frac{(x+2)}{2}$$

$$y-2 = \frac{x+2}{2}$$

or

$$y-2 = -\frac{(x+2)}{2}$$

$$2y-4 = x+2$$

or

$$2(y-2) = -x-2$$

$$\underline{x-2y+6=0}$$

$$2y-4 = -x-2$$

$$x+2y-2=0$$

pair 

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y-z-plane $x=0$

$$(y-z)^2 = z \Rightarrow \text{parabola}$$

x-z-plane $y=0$

$$(-z)^2 - \left(\frac{x+z}{4}\right)^2 = z$$

$$4 - \left(\frac{x+z}{4}\right)^2 = z$$

$$\left(\frac{x+z}{4}\right)^2 = 4-z \Rightarrow (x+z)^2 = 4(4-z)$$

Intercepts

x-intercept $y=0, z=0$

$$(0-2)^2 - \frac{(x+2)^2}{4} = 0 \Rightarrow 4 - \frac{(x+2)^2}{4} = 0$$

$$\frac{(x+2)^2}{4} = 4 \Rightarrow (x+2)^2 = 16 \Rightarrow x+2 = \pm 4$$

$$x+2=4 \quad \text{or} \quad x+2=-4$$

$$x=2$$

$$\text{or } x=-6$$

$(2, 0, 0)$ and $(-6, 0, 0)$

y-intercept $x=0, z=0$

$$(y-2)^2 - \frac{(0+4)^2}{4} = 0$$

$$(y-2)^2 - \frac{4}{1} = 0$$

$$(y-2)^2 = 1 \Rightarrow y-2 = \pm 1$$

$$y-2 = 1 \quad \text{or} \quad y-2 = -1$$

$$y = 3$$

$$y = 1$$

$(0, 3, 0)$ and $(0, 1, 0)$

Activate
Go to Settings

z-intercept $x=0, y=0$

$$(0-2)^2 - (0 \pm 2)^2 = z$$

$$4 - \frac{4}{4} = z$$

$$\Rightarrow z = 4 - 1$$

$$z = 3$$

$(0, 0, 3)$

Activate V
Go to Setting

Hyperboloid of one sheet

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} - \frac{(z-l)^2}{c^2} = 1$$

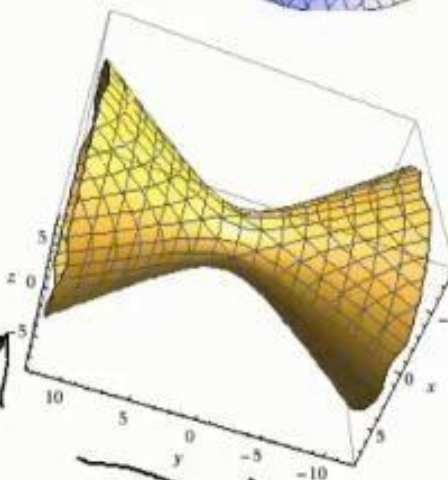
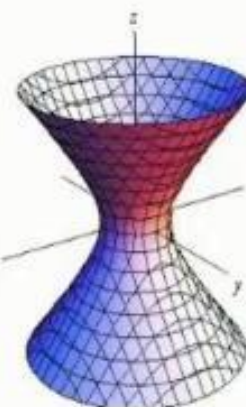
Centre (h, k, l)

if Centre $(0, 0, 0)$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \Rightarrow \text{along } z\text{-axis}$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \Rightarrow \text{along } y\text{-axis}$$

$$-\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \Rightarrow \text{along } x\text{-axis}$$



Activate V
Go to Setting

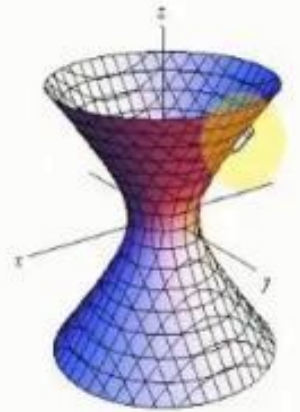
Symmetry

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

$$f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} - 1 = 0$$

(i) xy-plane $f(x, y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{(-z)^2}{c^2} - 1 = f(x, y, z) = 0$
 $\Rightarrow$ Symmetrical

(ii) yz-plane $f(-x, y, z) = \frac{(-x)^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} - 1 = f(x, y, z) = 0$
 $\Rightarrow$ Symmetrical



Activate \n
Go to Setting

(iii) xz-plane

$$f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} - 1 = f(x, y, z)$$

$\Rightarrow$ Symmetrical

(iv) x-axis

$$f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} - 1 = f(x, y, z) = 0$$

$\Rightarrow$ Symmetrical

(v) y-axis

$$f(-x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} - 1 = f(x, y, z) = 0$$

$\Rightarrow$ Symmetrical

Activate
Go to Setting

$\Rightarrow$ Symmetrical

(v) y-axis $f(-x, y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} - 1 = f(x, y, z) = 0$
 $\Rightarrow$ Symmetrical

(vi) z-axis

$$f(-x, -y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} - 1 = f(x, y, z) = 0$$

$\Rightarrow$ Symmetrical

(vii) origin $f(-x, -y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} - 1 = f(x, y, z) = 0$
 $\Rightarrow$ Symmetrical

Activate V
Go to Setting

Traces

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

(i) xy-plane put $z=0$

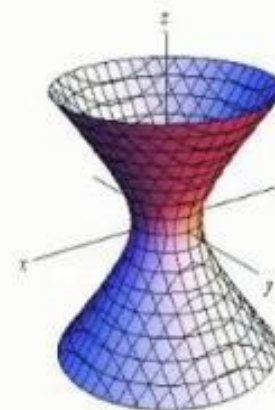
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \text{ellipse}$$

(ii) xz-plane put $y=0$

$$\frac{x^2}{a^2} - \frac{z^2}{c^2} = 1 \Rightarrow \text{Hyperbola}$$

(iii) yz-plane put $x=0$

$$\frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \Rightarrow \text{Hyperbola.}$$



Activate V
Go to Setting

Intercepts

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

(i) x-intercept put $y=0$, $z=0$

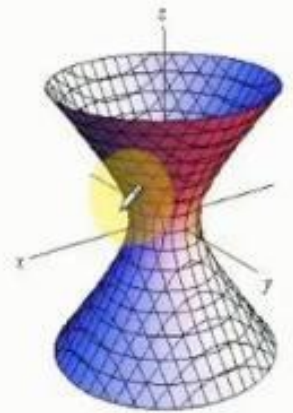
$$\frac{x^2}{a^2} = 1 \Rightarrow x^2 = a^2 \Rightarrow x = \pm a$$

$(a, 0, 0)$ and $(-a, 0, 0)$

(ii) y-intercept put $x=0$, $z=0$

$$\frac{y^2}{b^2} = 1 \Rightarrow y^2 = b^2 \Rightarrow y = \pm b$$

$(0, b, 0)$ and $(0, -b, 0)$



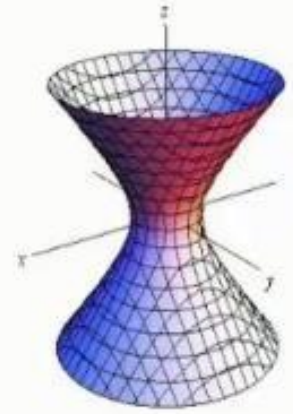
(ii) z intercept $x=0, y=0$

$$-\frac{z^2}{c^2} = 1 \Rightarrow z^2 = -c^2 \Rightarrow \text{No real sol.}$$

$\Rightarrow$ No z -intercept.

Boundary

not bounded



Example Hyperboloid of One sheet

Identify the surface

$$4x^2 + y^2 - 4z^2 - 16x - 4y + 8z + 12 = 0$$

and find its Centre and intercepts

Sol

$$4x^2 + y^2 - 4z^2 - 16x - 4y + 8z + 12 = 0$$

$$4x^2 - 16x + y^2 - 4y - 4z^2 + 8z = -12$$

$$4(x^2 - 4x + 4) + (y^2 - 4y + 4) - 4(z^2 - 2z + 1) = -12 + 16 + 4 - 4$$

$$4(x-2)^2 + (y-1)^2 - 4(z-1)^2 = 4$$

$$(x-2)^2 + \frac{(y-1)^2}{4} - (z-1)^2 = 1$$

Activate V
Go to Setting

$$(x-2)^2 + \frac{(y-1)^2}{4} - (z-1)^2 = 1$$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} - \frac{(z-l)^2}{c^2} = 1$$

⇒ Hyperboloid of one sheet
centre (2, 1, 1)

Activate V
Go to Setting

Traces

$$4x^2 + y^2 - 4z^2 - 16x - 4y + 8z + 12 = 0$$

(i) x-intercept: Put $y=0, z=0$

$$4x^2 - 16x + 12 = 0 \Rightarrow x^2 - 4x + 3 = 0 \Rightarrow (x-1)(x-3) = 0$$

$$x-1=0 \quad \text{or} \quad x-3=0$$

$$x=1 \quad \text{or} \quad x=3$$

$$(1, 0, 0) \quad \text{and} \quad (3, 0, 0)$$

(ii) y-intercept Put $x=0, z=0$

$$y^2 - 4y + 12 = 0 \Rightarrow \text{No solution}$$

No y-intercept

Activate V
Go to Setting

(iii) z-intercept $x=0, y=0$

$$-4z^2 + 8z + 12 = 0$$

$$z^2 - 2z - 3 = 0$$

$$(z-3)(z+1) = 0$$

$$z-3=0 \quad z+1=0$$

$$z=3 \quad z=-1$$

$(0, 0, 3)$ and $(0, 0, -1)$

Activate
Go to Settings

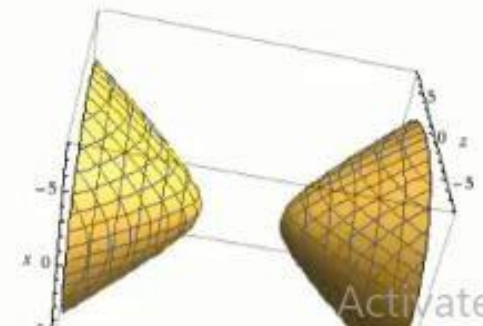
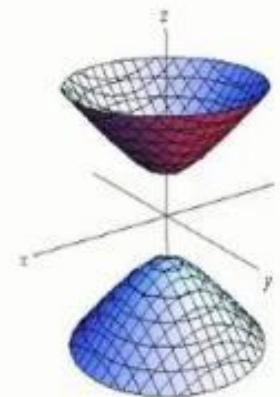
Hyperboloid of Two sheets

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} - \frac{(z-l)^2}{c^2} = -1$$

Centre (h, k, l)

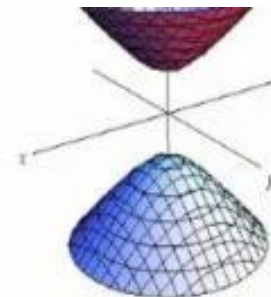
Centre $(0, 0, 0)$ $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$ $\longrightarrow$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1$$



Centre (h, k, l)
 Centre $(0, 0, 0)$

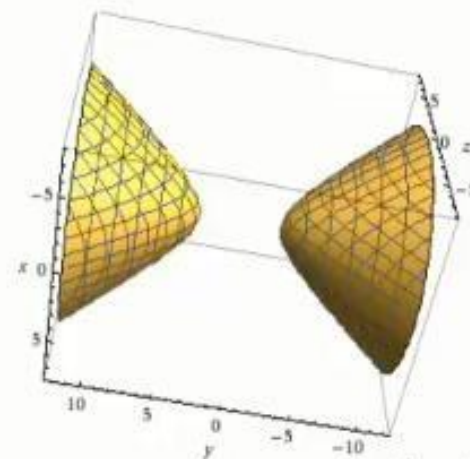
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1 \longrightarrow$$



$$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1$$

Centre $(0, 0, 0)$

$$-\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = -1$$



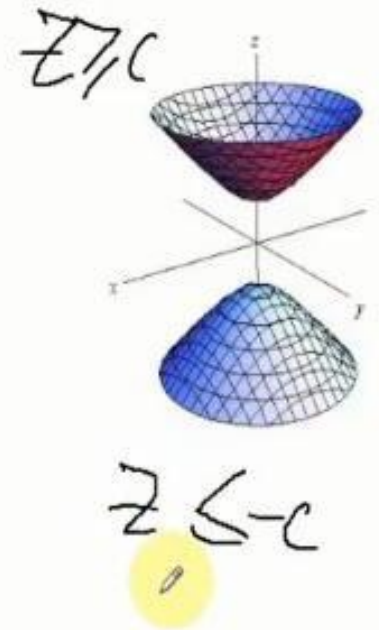
Activate
Go to Setting

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2} - 1$$

$$\text{i.e. } \frac{z^2}{c^2} - 1 > 0 \text{ or } \frac{z^2}{c^2} > 1$$

$$\text{or } z^2 > c^2 \text{ or } z < -c \text{ or } z > c$$



Symmetry $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$

$$f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} + 1 = 0 \rightarrow \textcircled{1}$$

(i) xy-plane $f(x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} + 1 = f(x, y, 0) = 0$

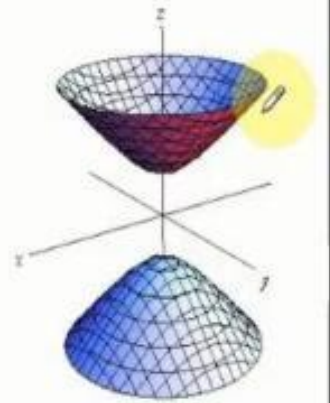
$\Rightarrow$ Symmetrical

(ii) xz-plane $f(x, -y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} + 1 = f(x, y, z) = 0$

$\Rightarrow$ Symmetrical xz-plane

(iii) yz-plane $f(-x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} + 1 = f(x, y, z) = 0$

Symmetrical



Activate V
Go to Setting

(iv) x-axis

$$f(x, -y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} + 1 = 0 = f(x, y, z)$$

Symmetrical

(v) y-axis $f(-x, y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} + 1 = f(x, y, z) = 0$

Symmetrical

(vi) z-axis $f(-x, -y, z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} + 1 = f(x, y, z) = 0$

$\Rightarrow$ Symmetrical

(vii) origin $f(-x, -y, -z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} + 1 = f(x, y, z) = 0$

Traces

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$

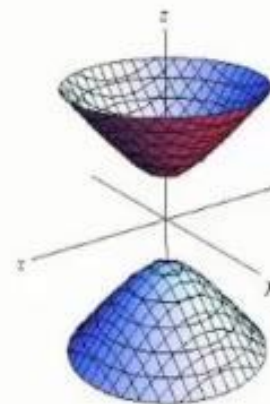
(i) xz-plane put $y=0$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = -1 \Rightarrow \text{No solution}$$

$\Rightarrow$ No trace

(ii) yz-plane put $x=0$

$$\frac{y^2}{b^2} - \frac{z^2}{c^2} = -1 \Rightarrow \frac{z^2}{c^2} - \frac{y^2}{b^2} = 1 \Rightarrow \text{Hyperbola}$$



(ii) yz-plane put $x=0$

$$(iii) \text{ } xz \text{ plane } y=0 \quad \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1 \Rightarrow \frac{z^2}{c^2} - \frac{y^2}{b^2} = 1 \Rightarrow \text{Hyperbola}$$

$$\frac{x^2}{a^2} - \frac{z^2}{c^2} = -1 \Rightarrow \frac{z^2}{c^2} - \frac{x^2}{a^2} = 1 \Rightarrow \text{Hyperb}$$

Intercepts

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$

(i) x-intercept put $y=0, z=0$

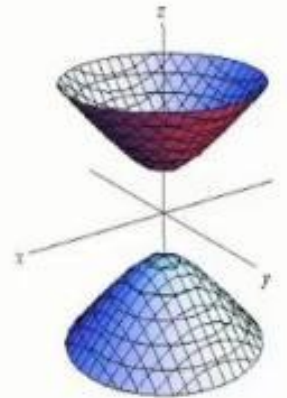
$$\frac{x^2}{a^2} = -1 \Rightarrow x^2 = -a^2 \Rightarrow \text{No solution}$$

$\Rightarrow$ No x-intercept

(ii) y-intercept $x=0, z=0$

$$\frac{y^2}{b^2} = -1 \Rightarrow y^2 = -b^2 \Rightarrow \text{No y-intercept}$$

(iii) z-intercept $x=0, y=0$ $-\frac{z^2}{c^2} = -1 \Rightarrow z^2 = c^2 \Rightarrow z = \pm c$
 $(0, 0, c)$ and $(0, 0, -c)$



Example Hyperboloid of Two sheets

Identify the surface

$$9x^2 + 9y^2 - z^2 - 18x - 18y + 2z + 26 = 0$$

find its centre and traces in coordinate planes

Solution

$$9x^2 + 9y^2 - z^2 - 18x - 18y + 2z + 26 = 0$$

$$9x^2 - 18x + 9y^2 - 18y - z^2 + 2z = -26$$

$$9(x^2 - 2x + 1) + 9(y^2 - 2y + 1) - (z^2 - 2z + 1) = -26 + 9 + 9 - 1$$

$$9(x-1)^2 + 9(y-1)^2 - (z-1)^2 = -9$$

Activate V
Go to Setting

$$9(x-1)^2 + 9(y-1)^2 - (z-1)^2 = -9$$

$$(x-1)^2 + (y-1)^2 - \frac{(z-1)^2}{9} = -1$$

$$\frac{(x-1)^2}{a^2} + \frac{(y-1)^2}{b^2} - \frac{(z-1)^2}{c^2} = -1$$

Hyperboloid of two sheets

$C(1,1,1)$

Activate V
Go to Setting

Traces $(x-1)^2 + (y-1)^2 - \frac{(z-1)^2}{9} = -1$

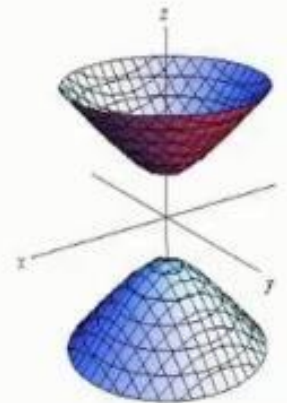
(i) xy-plane $z=0$

$$(x-1)^2 + (y-1)^2 - \frac{1}{9} = -1$$

$$(x-1)^2 + (y-1)^2 = -1 + \frac{1}{9} = -\frac{8}{9}$$

$\Rightarrow$ No solution

$\Rightarrow$ No trace



(i) yz-plane $x=0$ $1 + (y-1)^2 - \frac{(z-1)^2}{9} = -1$
 $(y-1)^2 - \frac{(z-1)^2}{9} = -2$
 $\frac{(z-1)^2}{9} - (y-1)^2 = 2 \Rightarrow \text{Hyperbola}$

(ii) xz-plane $y=0$
 $(x-1)^2 + 1 - \frac{(z-1)^2}{9} = -1 \Rightarrow (x-1)^2 - \frac{(z-1)^2}{9} = -2$
 $\frac{(z-1)^2}{9} - (x-1)^2 = 2 \Rightarrow \text{Hyperbola}$

Spherical Trigonometry

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Main Branches of Mathematics

- **Arithmetic**
- **Algebra**
- **Geometry**

Geometry

- The mathematical study of the properties, measurement, and relationships of points, circles, lines, planes, surfaces, angles etc.



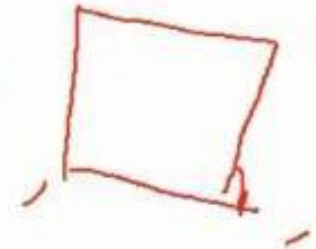
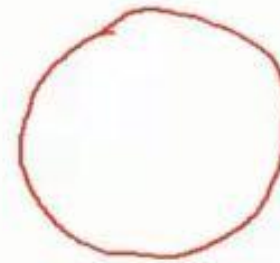
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Types of Geometry

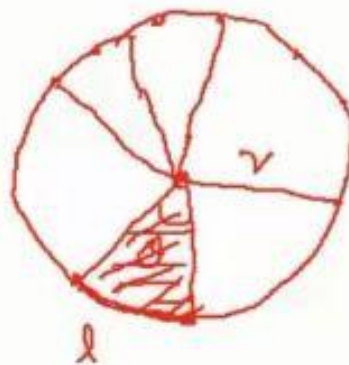
- Plane geometry
- Solid geometry
- Spherical geometry



plane geometry :



Example



$$\underline{C = 2\pi r}$$

$$\underline{A = \pi r^2}$$

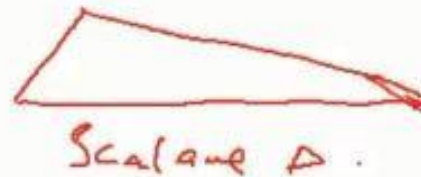
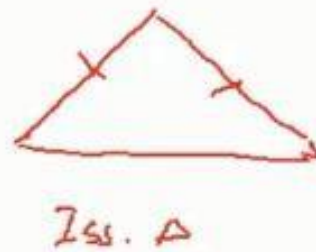
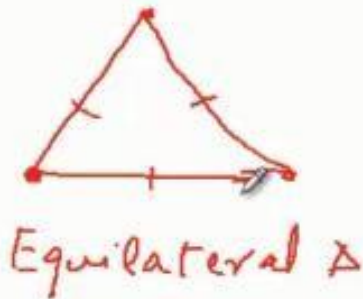
$$\underline{l = r\theta}$$

$$\underline{A = \frac{1}{2} r^2 \theta}$$

Activate V

Trigonometry

- Trigonometry is a branch of geometry that studies relationships involving lengths and angles of triangles.



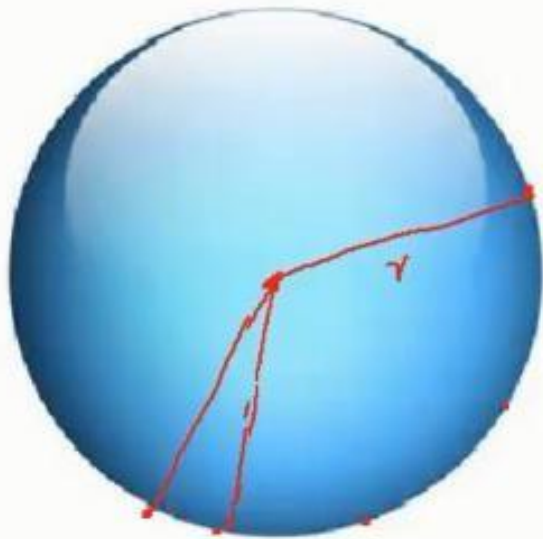
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Solid geometry

- It is the study about three dimensional objects like spheres, cubes, prisms, cylinders etc.

Example





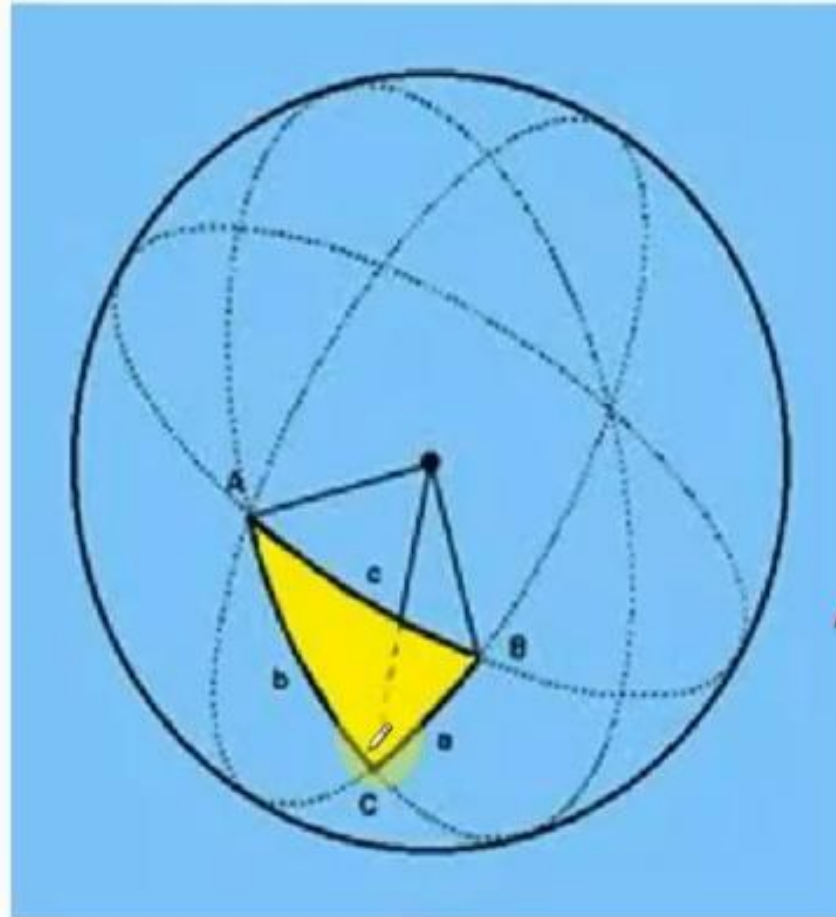
$$A = 4\pi r^2$$

$$V = \frac{4}{3}\pi r^3$$





The Cosine Formula



Join VU Group: <https://chat.whatsapp.com/KykRWALOWY90S1VJ72HWsM>

Activate
Go to S

The Lagrange's identity

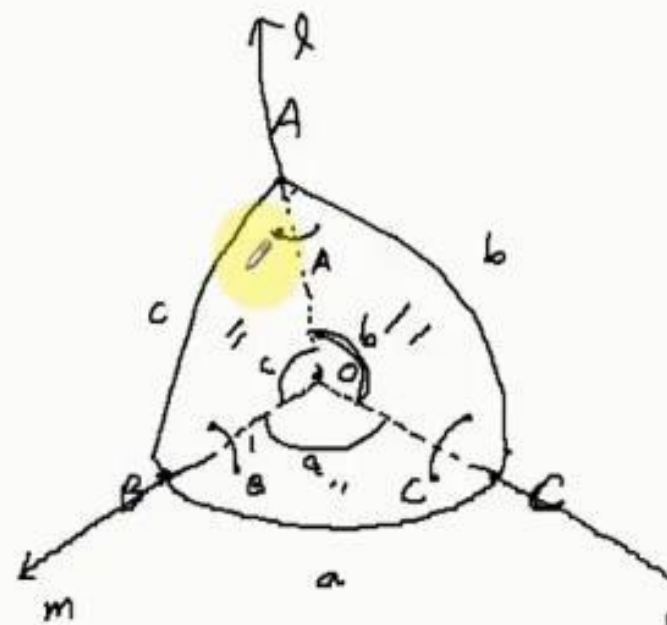
$$(l \times m) \cdot (l \times n) = \begin{vmatrix} l \cdot l & l \cdot n \\ m \cdot l & m \cdot n \end{vmatrix} \rightarrow (1)$$

Take $\begin{vmatrix} l \cdot l & l \cdot n \\ m \cdot l & m \cdot n \end{vmatrix} = \begin{vmatrix} |l||l|\cos 0 & |l||n|\cos b \\ |m||l|\cos c & |m||n|\cos a \end{vmatrix}$

$$\text{RHS} = \begin{vmatrix} 1 & \cos b \\ \cos c & \cos a \end{vmatrix} \quad \because |l| = |m| = |n| = 1$$

$$\text{RHS} = \cos a - \cos b \cos c \rightarrow (2)$$

$$\text{L.H.S} = (l \times m) \cdot (l \times n)$$



$$\begin{aligned} l &= r\theta \\ b &= (1)\theta \\ \theta &= b \end{aligned}$$

The Lagrange's identity

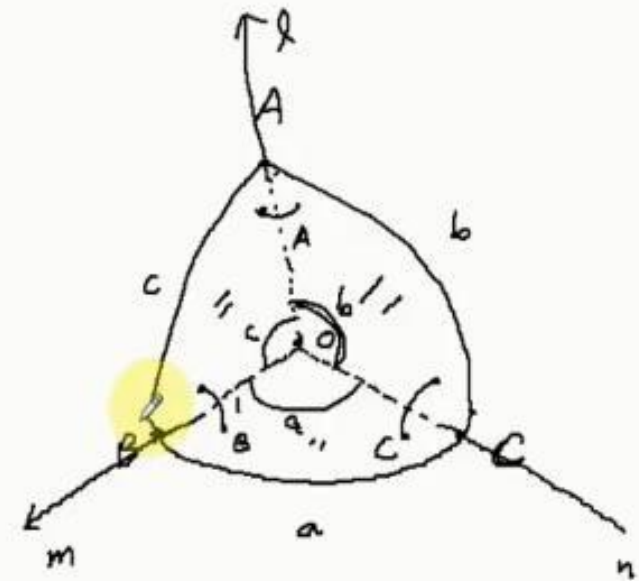
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$$\text{L.H.S} = (l \times m) \cdot (l \times n)$$



$$\begin{aligned} l &= r\theta \\ b &= (1)\theta \\ a &= b \end{aligned}$$

Activate V
Go to Setting

$$\text{RHS} = \cos a - \cos b \cos c \rightarrow (2)$$

$$b = (1) \theta$$

$$c = b$$

$$\text{L.H.S} = (l \times m) \cdot (l \times n)$$

$$= |l \times m| |l \times n| \cos A$$

$$= |l| |m| \sin c \cdot |l| |n| \sin b \cdot \cos A$$

$$= \sin c \sin b \cos A \rightarrow (3)$$

From eqns 2 & 3

$$\cos a - \cos b \cos c = \sin c \sin b \cos A$$

$$\cos a = \cos b \cos c + \sin b \sin c \cos A$$

$$\text{RHS} = \cos a - \cos b \cos c \rightarrow (2)$$

$$b = (1) \theta$$

$$c = b$$

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From eqns 2 & 3

$$\cos a - \cos b \cos c = \sin c \sin b \cos A$$

$$\cos a = \cos b \cos c + \sin b \sin c \cos A$$

$$\cos b = \cos a \cos c + \sin a \sin c \cos B$$

$$\cos c = \cos a \cos b + \sin a \sin b \cos C$$

The Sine Formula

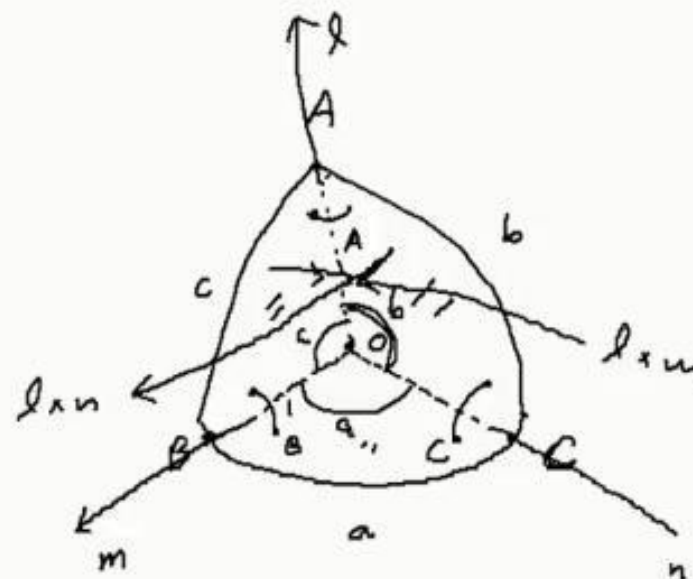
$$(l \times m) \times (l \times n) = [lmn]l - [lm l]n$$

$$(l \times m) \times (l \times n) = [lmn]l \quad \because [lm l] = 0$$

$\rightarrow \textcircled{1}$

$$\begin{aligned} \underline{(l \times m)} \times \underline{(l \times n)} &= |l \times m| |l \times n| \sin A \, l \\ &= |l| |m| \sin C \cdot |l| |n| \sin B \cdot \sin A \, l \end{aligned}$$

$$(l \times m) \times (l \times n) = \sin B \sin C \sin A \, l \quad \because |l| = |m| = |n| = 1$$



$$\begin{aligned} l &= r \cdot a \\ b &= 1 \cdot a \\ a &= b \end{aligned}$$

Activate V
Go to Settings

$$a = b$$

$$\text{Eq ①} \Rightarrow \sin b \sin c \sin A = [\Delta mn]$$

$$\Rightarrow \sin b \sin c \sin A = [\Delta mn] \rightarrow \text{②}$$

$$\sin a \sin b \sin C = [\Delta mn] \rightarrow \text{③}$$

$$\sin a \sin c \sin B = [\Delta mn] \rightarrow \text{④}$$

$$\Rightarrow \sin b \sin c \sin A = \sin a \sin b \sin C = \sin a \sin c \sin B$$

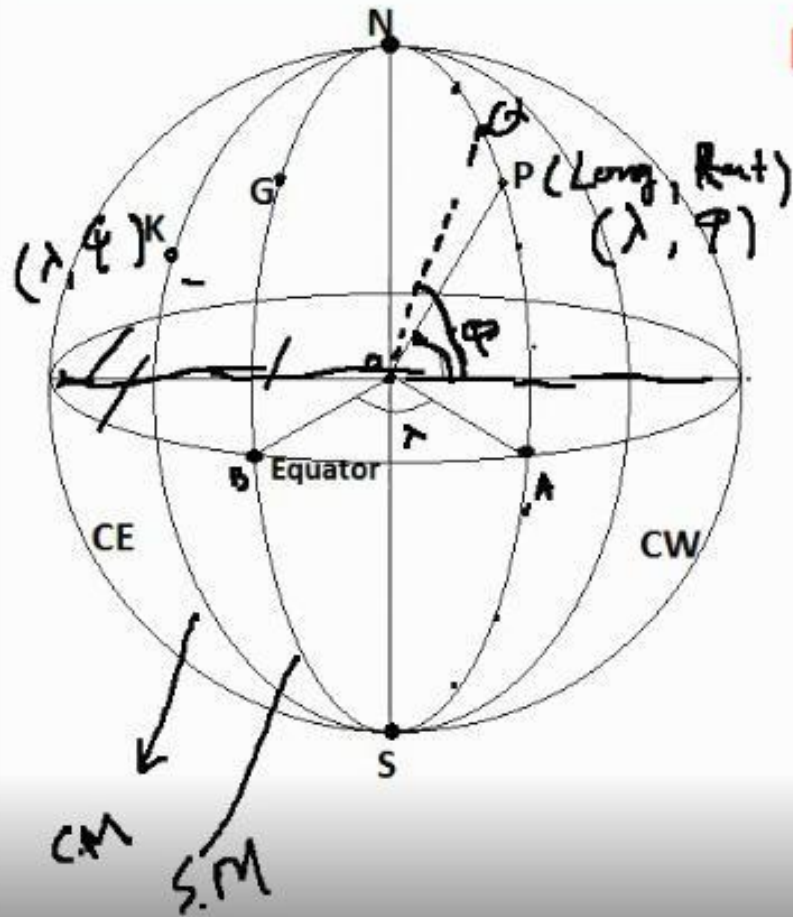
Dividing by $\sin a \sin b \sin c$

$$\Rightarrow \boxed{\frac{\sin A}{\sin a} = \frac{\sin C}{\sin c} = \frac{\sin B}{\sin b}}$$

The Sine Formula.



Activate V
Go to Setting



Latitude

λ_0 = Longitude of Khana-e-Kaba

= $39^{\circ} 49.2' E$ ✓

ϕ_0 = Latitude of Khana-e-Kaba

= $21^{\circ} 25.2' N$ ✓

==

Method to find the Direction of Qibla

λ = longitude of any place on the surface of earth

φ = latitude " " " " " " " "

$\lambda_0 = 39^\circ 49.2' E$ = longitude of Khan-e-Kaba

$\varphi_0 = 21^\circ 25.2' N$ = latitude " " " "

$$p = \frac{\sin \varphi}{\tan \lambda}, \quad q = \frac{\cos \varphi \tan \varphi_0}{\sin \lambda}$$

$$\tan i = p - q$$

$$i = \tan^{-1}(p - q) \quad \checkmark$$

The classical longitude l of a place can be found by the following way.

1. If $\lambda_0 < \lambda^0 E \leq 180^0$ then $l = (\lambda - \lambda_0)^0 \underline{CE}$
2. If $0^0 \leq \lambda^0 E < \lambda_0$ then $l = (\lambda_0 - \lambda)^0 CW$ ✓
3. If $0 < \lambda^0 W < 180^0 - \lambda_0$ then $l = (\lambda_0 + \lambda)^0 CW$ ✓



$$\lambda = 73^{\circ} 4' E, \quad \varphi = 33^{\circ} 40' N$$

$$\lambda = 73^{\circ} 4' E, \quad \varphi = 33^{\circ} 40' N$$

Longitude of Islamabad = $\lambda = 73^{\circ} 8' E$

Longitude of Islamabad = $\lambda = 73^{\circ} 8' E$

Latitude " " " = $\varphi = 33^{\circ}40'N$

Latitude " " " = $\varphi = 33^{\circ}40'N$

Longitude of Kharan-e-Kabla = $\lambda = 39^{\circ}49.2' E$

Longitude of Kharan-e-Kabla = $\lambda = 39^{\circ}49.2' E$

Latitude " " " = $\phi_0 = 21^\circ 25.2' N$

Latitude " " " = $\phi_0 = 21^\circ 25.2' N$

Longitude of Khana-e-Kaba = $\lambda = 39^{\circ}49.2' E$

Latitude " " " = $\phi_0 = 21^{\circ}25.2' N$

$$p = \frac{\sin \phi}{\tan l}$$

$$l = \lambda - \lambda_0 = 73^{\circ}8' - 39^{\circ}49.2'$$

$$l = 33^{\circ}18.8' (CE)$$

$$p = \frac{\sin 33^{\circ}40'}{\tan 33^{\circ}18.8'} = 0.8435$$

$$q = \frac{\cos \phi \tan \phi_0}{\sin l} = \frac{\cos 33^{\circ}40' \tan 21^{\circ}25.2'}{\sin 33^{\circ}18.8'}$$

$$q = 0.1793$$

$$q = \frac{\cos \phi \tan \phi_0}{\sin \lambda} = \frac{\cos 33^\circ 40' \tan 21^\circ 25.2'}{\sin 33^\circ 18.8'}$$

$$q = 0.1793$$

$$\text{As } \tan i = p - q$$

$$= 0.8635 - 0.1793$$

$$= 0.6642$$

$$i = \tan^{-1}(0.6642) = 33^\circ 35.5' \quad \text{South of west } \checkmark$$